

A library of ideas

Around 1220, Mongol forces invaded Nasir's homeland, led by their ruthless ruler, Genghis Khan. Nasir found safety at the fortress of Alamut, in modern-day Iran. He liked it there because it had a brilliant library where he could study. When Alamut was destroyed by the Mongols in 1256, Nasir was appointed their scientific adviser. He persuaded Genghis Khan's grandson, Hulegu, to build a huge observatory—a place where astronomers study space—in modern-day Azerbaijan.

Persian and Chinese astronomers worked side-by-side to build and operate instruments for measuring the movements of planets and stars. Nasir invented many of these instruments himself, too. The observatory became an important center of learning, and was complete with another fantastic library.

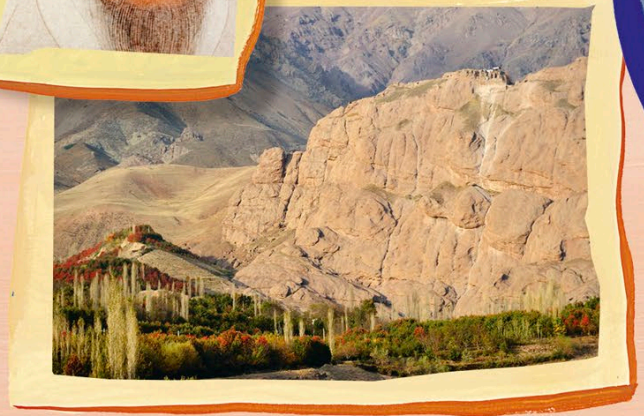
Ever-changing animals

Nasir observed Earth just as carefully as space. He came up with a theory to explain how animals evolve, or change, over time. Adaptations he described include the long horns of the Persian gazelle and the sharp claws of a falcon.

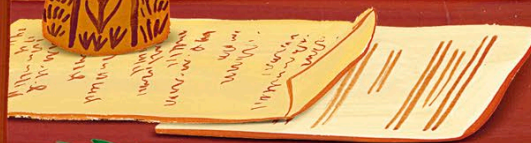
Genghis Khan



At the fortress of Alamut, Nasir wrote about many topics, including ethics—the study of what is right and wrong.



Nasir created more than 160 books and papers, including important texts on astronomy, mathematics, and biology. For example, he wrote about the importance of trigonometry (the study of triangles) in answering huge mathematical problems. His questions and ideas inspired other scientists for centuries to come—in Islamic lands and far beyond.



Nasir's work may have inspired other scientists, such as Nicolaus Copernicus, hundreds of years later!